ST4248 Term Project

Movie Box Office Revenue EDA

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In this paper, we examine three factors that may influence a film's financial success. First, we study how box office revenue differs across genres and apply Lasso regression to identify which genres are most associated with high revenue. Second, we explore the relationship between IMDb ratings and revenue, and consider how viewer engagement, measured by number of votes, may enhance this relationship. Finally, we analyze how the original language of a film relates to its dependence on international audiences, grouping languages by production context to identify broader patterns. Through these analyses, we aim to uncover key traits that help explain variation in global movie earnings.

Data Preprocessing

Our dataset consists of 5,000 unique movies, each containing information on worldwide box office revenue, domestic and foreign revenue shares, release year, associated genres (each movie has 1 to 5 genres), TMDB rating, vote count, original language, and filming locations. (Source: https://www.kaggle.com/datasets/aditya126/movies-box-office-dataset-2000-2024/data)

Handling Missing Values

Several variables contain missing values, all originating from the same set of movies. To maintain data consistency, we exclude these movies, reducing the dataset to 4,797 unique entries.

Revenue Distribution & Transformation

A histogram of total revenue reveals a highly right-skewed distribution (see Figure A1), as most movies earn modest amounts compared to blockbusters. To mitigate skewness in our analysis, we apply a logarithmic transformation to revenue (see Figure A2).

Genre Encoding

Since each movie is associated with multiple genres stored as a list, we apply one-hot encoding to convert them into separate binary variables, enabling effective regression analysis.

Rating & Vote Count Filtering

Ratings are originally stored as character values and are converted to numerical form. We also identify some movies with a rating of 0 due to having no votes. To prevent skewed results, we exclude movies with fewer than 30 votes, ensuring a more reliable rating variable. Vote count is also highly right skewed (see Figure A3), and thus we shall apply a logarithmic transformation to vote count (see Figure A4).

Original Language Cleaning

The original language data is provided in the form of language codes, which may not be intuitive for analysis. Additionally, some countries have multiple languages represented by different codes, resulting in limited data for certain language categories. To address this, we combine languages commonly spoken in the same country or within countries that share similar linguistic and cultural contexts, such as the Nordic countries. We also excluded edge cases where films are released domestic-only or international-only. Finally, we exclude any combined languages that have fewer than 5 data points to maintain the robustness and reliability of our results.

With these preprocessing steps completed, our dataset is now ready for analysis.

Part I: Revenue vs Genre

In this section, we will explore how worldwide box office revenue differs across various movie genres. Our goal is to examine whether certain genres tend to generate higher revenues than others, and if there are notable disparities in earnings between them. By analyzing the distribution of revenue within each genre, we aim to gain insight into general trends in box office performance, without implying value judgments on the artistic or cultural merit of different genres.

We will first look at the boxplot of the log revenue across genres.

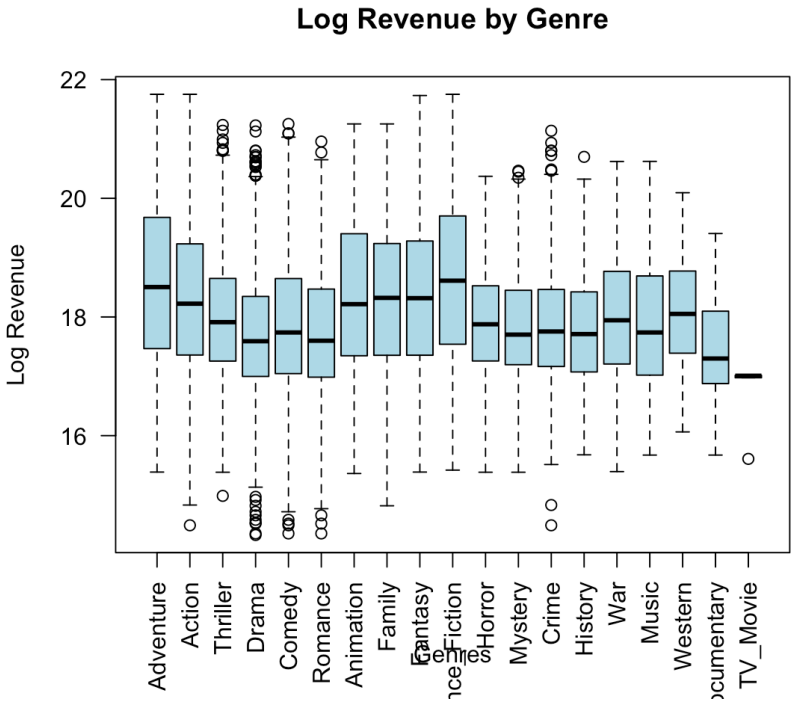


Figure 1: Boxplot of log worldwide revenue across genres

We observe that Science Fiction films, on average, earn the highest worldwide revenue, followed closely by Adventure, Fantasy, and Family genres. This is unsurprising, as these genres often overlap in popular films (see Figure B1). In contrast, TV Movies and Documentaries tend to generate lower revenue compared to other genres. Additionally, several genres, including Thriller, Drama, Comedy, Romance, Mystery, and Crime, exhibit a significant number of outliers. These outliers represent movies that either performed poorly or became blockbusters within their respective genres.

Next, we will fit Lasso regression with log worldwide revenue as the response against all the genres. We choose to employ Lasso regression over Kruskal-Wallis test or ANOVA due to its advantages in interpretability. While all 3 methods are able provide insights into whether there are significant differences in revenue distribution across genres, only

Lasso quantifies the relationship between the genres and the response, which is the log worldwide revenue.

Lasso regression coefficients									
Adventure	0.43	Comedy	-0.08	Fantasy	0.13	Crime	NA	Western	NA
Action	0.26	Romance	NA	Science Fiction	0.43	History	NA	Documentary	-0.31
Thriller	NA	Animation	NA	Horror	NA	War	0.07	TV Movie	-0.64
Drama	-0.19	Family	0.22	Mystery	NA	Music	0.10	Intercept	17.78

Table 1: Coefficients from Lasso Regression

Interpretation of the Coefficients:

- Positive Coefficients: Genres such as Science Fiction (0.43), Adventure (0.43), Family (0.22), and Action (0.26) have positive coefficients, indicating that these genres are associated with higher average worldwide revenue.
- Negative Coefficients: Documentary (-0.31) and TV Movie (-0.64) indicate lower average revenue compared to the baseline.
- Zero Coefficients (NA): Genres like Thriller, Romance, Animation, Horror, Mystery, Crime, History, and Western have been shrunk to zero, suggesting they do not significantly contribute to predicting revenue.

Overall, the Lasso regression analysis reveals which genres are most influential in predicting worldwide box office revenue while providing insights into the financial implications of different genre choices in filmmaking.

Part 2: Revenue vs Rating

In this section, we explore the relationship between a movie's rating and its worldwide revenue. While it is often assumed that well-rated movies perform better at the box office, the reality may be more nuanced. We aim to investigate whether higher ratings are associated with higher revenue, and whether this relationship holds after considering the number of votes a movie receives.

To start with, we plot the scatterplot of movie ratings against log(Worldwide Revenue). The plot revealed a diffused circular, concentrated around the rating range of 5 to 8 and log revenue between 16 to 20. This suggests that rating alone is a weak predictor of revenue, with R^2 value of just 0.02 when fitted into a linear model.

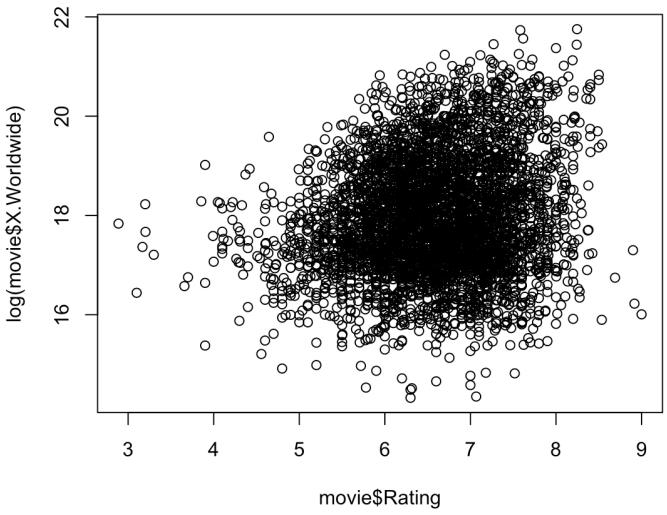


Figure 2: Scatterplot of log(Worldwide revenue) against Rating

To better understand this relationship, we introduce a new composite variable: Rating × log(Vote_Count). This variable reflects not only the quality of a movie as perceived by viewers but also how popular or widely watched it is. When we plot this new metric against log(Worldwide Revenue), a much clearer linear relationship emerges (with R^2 = 0.3633), indicating that both highly rated and widely reviewed movies tend to perform better financially.

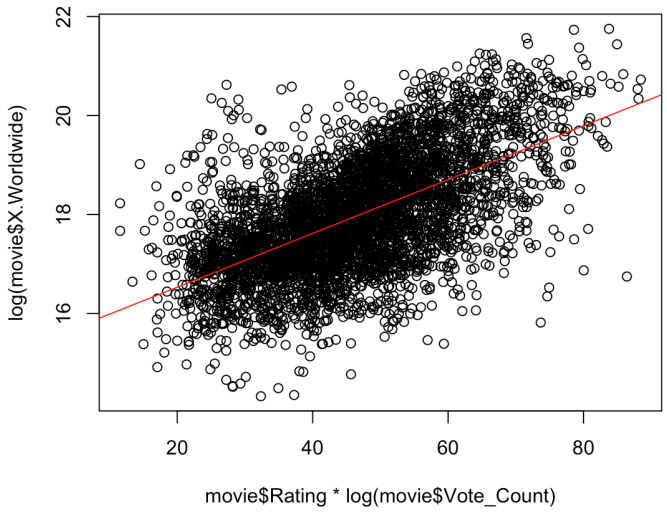


Figure 3: Scatterplot of log(Worldwide Revenue) against Rating x log(Vote Count)

Part 3: Film's language vs Foreign Revenue %

This section explores how films in different original languages vary in their reliance on international audiences. By analyzing the percentage of revenue earned from foreign markets, we aim to understand whether certain languages, and by extension, film industries, tend to perform better globally or remain more domestically focused.

While extreme cases of domestic-only or international-only releases have been excluded from the analysis, examining the general distribution of foreign revenue across languages still provides insight into how language influences a film's global appeal, reach, and commercial strategy.

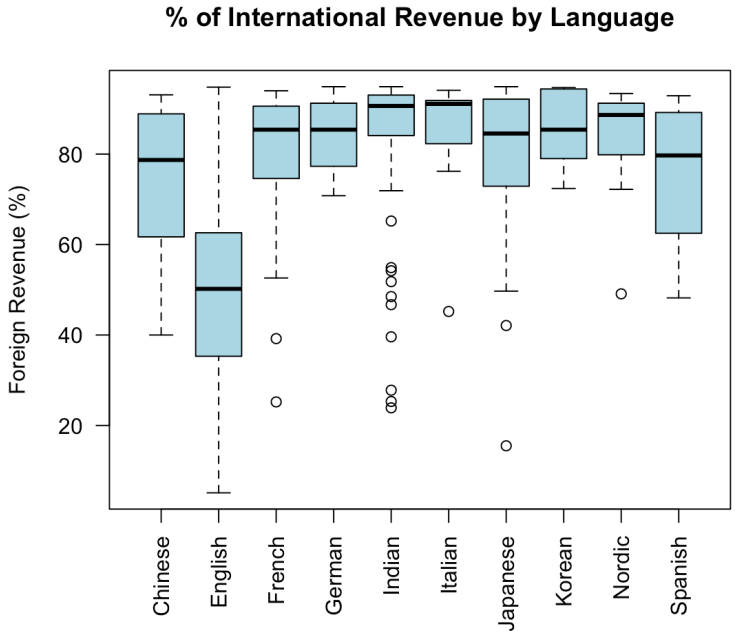


Figure 4: Boxplot of International Revenue percentage by Language

From Figure 4, we observe that most non-English films tend to rely heavily on international audiences for the bulk of their revenue. This is likely due to the smaller domestic markets in many countries, making global reach essential for commercial success. In contrast, English-language films display a much wider range in foreign revenue percentage. This variation reflects the diversity within the English film industry, ranging from small indie productions to global blockbusters and films specifically made for international markets.

The outliers in the data also offer interesting insights. Among French films, the lowest foreign revenue percentage belongs to a small domestic production, while the second lowest is a relatively popular film that likely performed better locally. For Indian films, those with lower foreign revenue shares may be deeply rooted in local culture or folklore, limiting their international appeal. The outliers in Italian and Nordic films likely represent titles that performed well domestically but failed to gain traction abroad. Finally, the two Japanese outliers appear to be smaller films intended primarily for local audiences.

Appendix A: Data Preprocessing

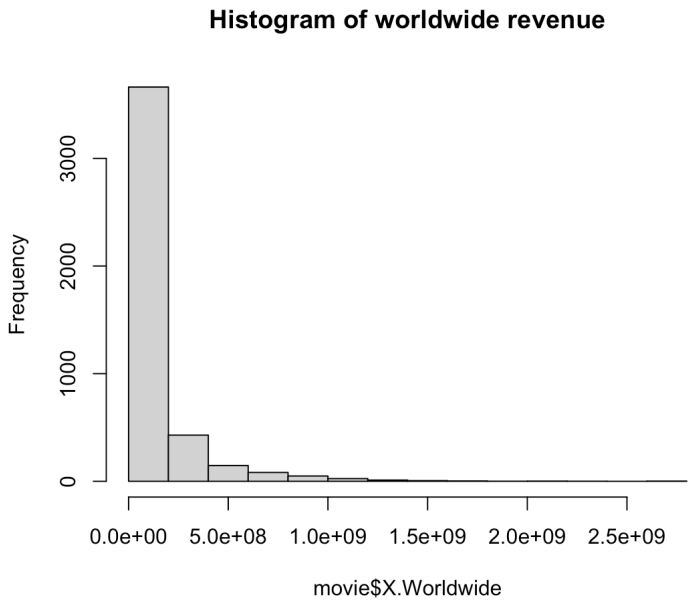


Figure A1: Histogram of worldwide revenue

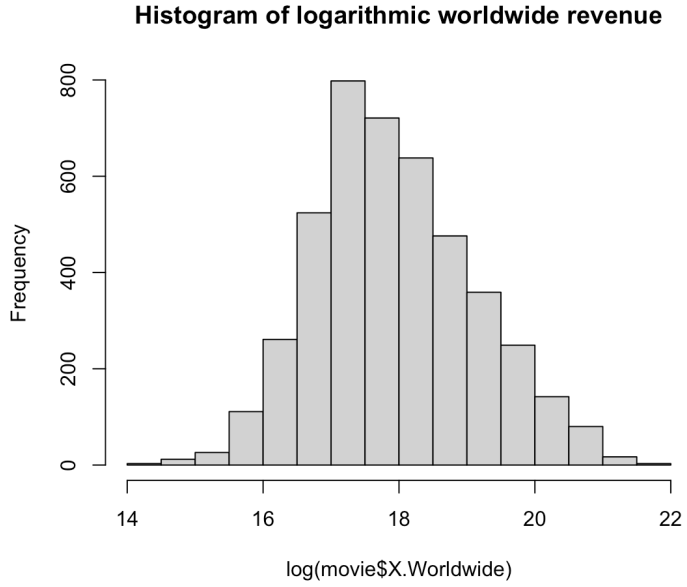


Figure A2: Histogram of logarithmic worldwide revenue

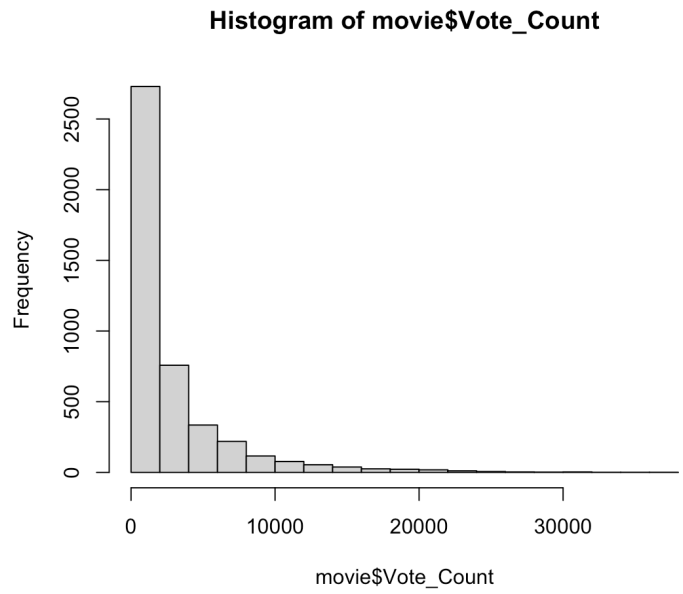


Figure A3: Histogram of TMDB vote count

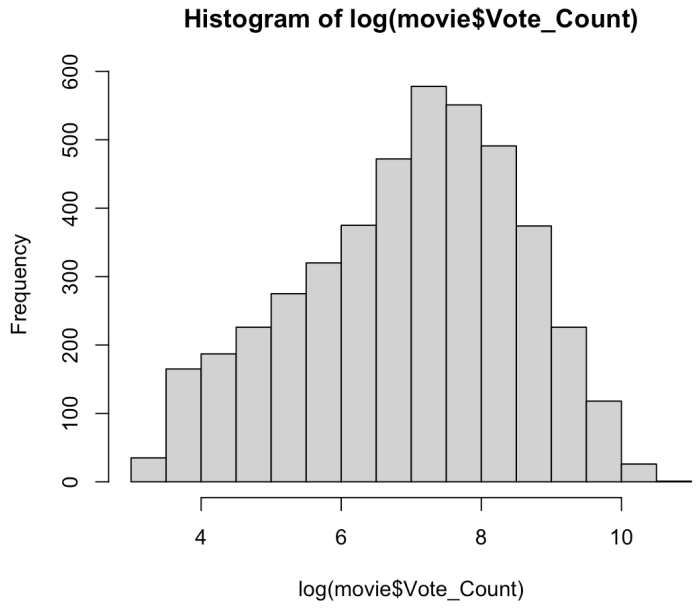


Figure A4: Histogram of log transformed vote count

Appendix B: Revenue vs Genre

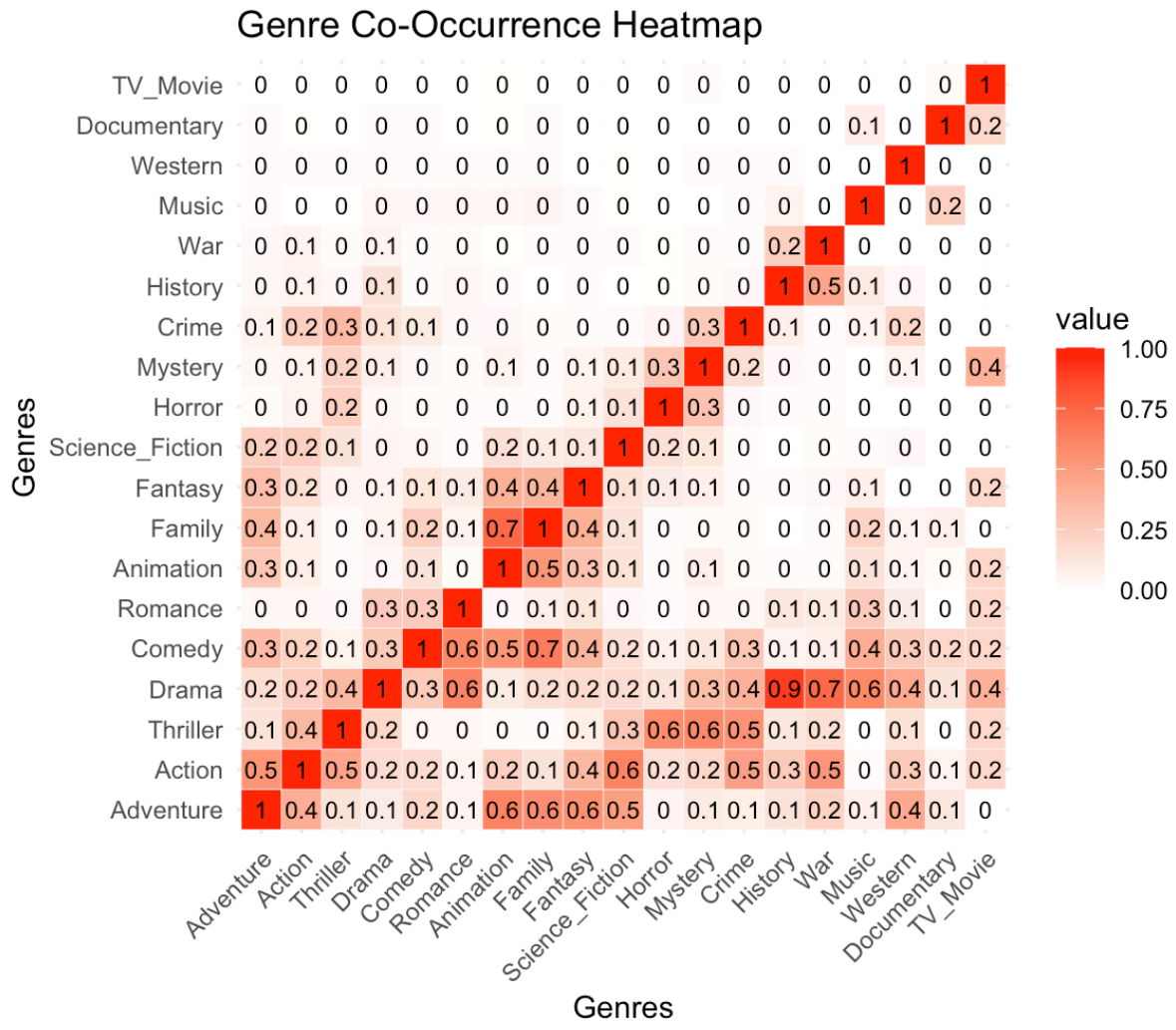


Figure B1: Genre Co-Occurrence Heatmap, showing how often does a genre overlap with another in a movie